DTU-Net: A Multi-Scale Dilated Transformer Network for Nonlinear Hyperspectral Unmixing

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***Abstract*—Transformers have shown significant success in hyperspectral unmixing (HU). However, challenges remain. Transformer-based unmixing networks, built on vision transformer (ViT) or Swin transformer, struggle to effectively capture essential multiscale and long-range spatial correlations. In addition, these networks predominantly rely on the linear mixing model (LLM), lacking the flexibility to accommodate scenarios with significant nonlinear effects. To address these limitations, we propose a multiscale dilated transformer-based unmixing network for nonlinear HU (DTU-Net). Its encoder integrates two branches: a spatial branch mainly employing multiscale dilated attention (MSDA) to uniquely capture intricate multiscale and long-range spatial correlations via adaptive receptive fields, and a spectral branch utilizing 3-D-convolutional neural networks (CNNs) with channel attention. This design enables comprehensive extraction as well as integration of multilevel spatial and spectral features. The decoder is specifically designed to accommodate both linear and nonlinear mixing. It explicitly models the polynomial post-nonlinear mixing model (PPNMM) by learning nonlinear coefficients as pixel-wise features, which enhances interpretability by directly reflecting the pixel-level nonlinear mixing strength. Experiments on synthetic, ray tracing, and real datasets validate the effectiveness of the proposed DTU-Net, demonstrating its superior performance compared to both PPNMM-derived and advanced unmixing networks. The code is available at: https://github.com/ChentongWang/DTU-Net**

***Index Terms*—Autoencoder (AE), polynomial post-nonlinear mixing model (PPNMM), spectral unmixing (SU), transformer.**

I. Introduction

HYPERSPECTRAL images (HSIs) provide detailed observations of a scene across hundreds of contiguous, narrow wavelength bands, enabling land-cover classification, anomaly detection, and other material-discrimination tasks [1]. While HSIs deliver rich spectral information, they often have limited spatial resolution. Consequently, each pixel often comprises a mixture of spectra from multiple ground cover materials. The distinct spectral signatures associated with these materials are termed endmembers, while their proportional contributions within a pixel are referred to as abundances. Spectral unmixing (SU) aims to identify these endmembers and estimate their corresponding abundances for each pixel [2].

Prior to SU, a mixing model is typically adopted to characterize the light scattering and interaction mechanisms among the constituent materials within a hyperspectral pixel. The linear mixing model (LMM) is widely used for its simplicity and interpretability. It assumes that the incident ray of light interacts with a single pure material before reaching the sensor, modeling each pixel as a linear combination of endmembers weighted by abundances [3]. However, the LMM often struggles when nonlinear effects are significant. To address this, various nonlinear unmixing models have been proposed. These include Hapke's model [4], kernel-based models [5], [6], [7], mismodeling-based models [8], [9], [10], [11], [12], bilinear mixing models (BMMs) [13], [14], [15], [16], and multilinear mixing models (MLMs) [17], [18], [19]. Hapke's model is a radiative transfer model that describes light interaction with intimate mixtures. However, it has high complexity and demands scene-specific parameters which are often unavailable [20]. Higher order models such as MLMs theoretically extend modeling capabilities, yet the spectral amplitudes of their higher order interactions are typically very low in practice, often making them indistinguishable from noise [16].

Bilinear mixing, a fundamental form of nonlinear mixing, occurs when photons successively interact with two distinct materials before reaching the sensor. This phenomenon is particularly significant in forested regions, where interactions between soil and vegetation are prominent [1]. Mathematically, BMMs represent each pixel as the sum of a linear component and second-order interaction terms. Depending on the constraints applied to the interaction coefficients, various bilinear models have been proposed [13], [14], [15], [16]. While many BMMs [13], [14], [15] require multiple parameters to describe pixel-wise nonlinearities, the polynomial post-nonlinear mixing model (PPNMM) [16] offers a more compact representation. It models polynomial nonlinearity with a single scalar b and notably generalizes the LMM, reverting to it when $b = 0$. To balance modeling accuracy and computational complexity, this article adopts the PPNMM as its primary mixing model.

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Recent advancements in deep learning have been increasingly applied to unmixing, with autoencoder (AE) as a prominent framework [20]. Early AEs typically operate under the LMM and perform single-pixel processing. The encoder maps observed pixels to low-dimensional representations, interpreted as abundances, while a fully connected (FC) decoder reconstructs the pixels, with its weights corresponding to the estimated endmember matrix [21], [22], [23]. To effectively capture the spectral–spatial structure of hyperspectral imagery, convolutional neural networks (CNNs) have been integrated into AE-based unmixing networks, primarily enhancing the encoder to enable implicit regularization of abundance estimation. For example, the cycle-consistency unmixing network (CyCU-Net) [24] uses 2-D convolutions to preserve spatial details, while CNNAEU [25] utilizes image patches to enable neighboring pixels to contribute to the reconstruction. CNNs have also been incorporated into nonlinear unmixing networks [26], [27], [28], [29], which further improve the modeling of spectral–spatial relationships. Nevertheless, despite their effectiveness in capturing local features, CNNs struggle to model long-range spatial dependencies within HSIs.

A. Attention in Transformer-Based Unmixing Networks

Precise unmixing requires handling the intrinsic spatial complexity of materials, ranging from fine textures to vast homogeneous regions, by leveraging multiscale features for local details and long-range dependencies for global context. Transformers have achieved broad success across domains by leveraging self-attention mechanisms [30]. A prominent variant, the vision transformer (ViT) [31], effectively models long-range dependencies in images by attending to global context and spatial relationships through multihead self-attention (MHSA). MHSA allows the model to jointly attend to information from different representation subspaces, providing a powerful mechanism for global feature extraction. In unmixing, deep transformer network (DeepTrans) [32] combines a CNN and ViT encoder with multihead self-patch attention to extract both local and global spatial features, while the work in [33] employs a ViT encoder with a CNN-based decoder that incorporates a dual-channel graph regularizer to model semantic and manifold structural information. In [34], a self-attention mechanism is directly used to estimate endmembers, abundances, and reconstruct images, a departure from prior works that only utilized attention for feature extraction. However, despite ViT's strengths in capturing global spatial features, its dense, uniform attention mechanism lacks adaptability to spatial scale variations and can be computationally expensive for high-resolution HSIs.

In response, recent unmixing works have explored more efficient and flexible attention strategies over ViT, primarily inspired by the Swin transformer [35]. These widely adopted strategies include window-based MHSA (W-MSA) mechanisms, notably the shifted window MHSA (SW-MSA) strategy [36], [37], [38], [39]. Examples include UnDAT [36], which employs a superpixel-guided W-MSA module where spatially homogeneous regions define attention windows for cross-window interaction. U-shaped transformer network using shifted windows (UST-Net) [37] integrates SW-MSA with downsampling and upsampling to capture multiscale spatial features, while Swin-hyperspectral unmixing (HU) [38] applies the same mechanism for multiscale spatial dependency modeling. In addition, a hybrid transformer–CNN encoder in [39] uses pixel embeddings for pixel-level window attention (WP-MSA), enabling nonlocal modeling while preserving spatial resolution for sparse unmixing.

While window-based attention mechanisms like W-MSA and SW-MSA significantly reduce computational complexity by confining self-attention to fixed-size local windows, this design introduces a key limitation. Their uniform receptive fields across all attention heads within an attention block fundamentally restrict the model's capacity for true multiscale feature extraction. Despite hierarchical context propagation facilitated by spatial shifts, this rigidity makes them less adaptable to the intricate and scale-variant spatial dependencies inherent in HSIs. However, accurately modeling multiscale and long-range dependencies remains challenging, yet crucial for precise unmixing of scenes with fine local heterogeneity and global structural consistency. Paralleling these spatial limitations, extracting discriminative spectral features is equally critical. Although recent transformer-based methods [34], [36] incorporate 1-D-CNNs or specific modules to model intragroup spectral relationships, they typically employ hand-crafted or fixed spectral grouping strategies, failing to adaptively recalibrate spectral channel importance and limiting sensitivity to subtle spectral features in complex scenarios. Another limitation across existing transformer-based unmixing networks [32], [33], [34], [36], [37], [38], [39] is their reliance on the linear mixing assumption, which severely restricts their applicability to real-world scenarios where nonlinear effects are significant.

B. Nonlinear Unmixing Networks

While many existing networks rely on the LMM, others have advanced to nonlinear unmixing by integrating established nonlinear physical models, typically within their decoder architectures [20]. For instance, HapkeCNN [40] targets intimate mixtures, adopting a quadratic term based on the Hapke model within its loss function to capture nonlinearity. Alternatively, 3-D-NAE [26] adapts to the linear-mixture/nonlinear-fluctuation mixing model, wherein the nonlinear relationship among the endmembers is implicitly characterized by FC layers [5]. Networks based on MLM-based models [29], [41] have been proposed to enhance the physical interpretability of the unmixing process, notably by learning transfer probabilities in the mixing model as features.

As a key category of nonlinear models, BMMs have been widely adopted in network design. For example, [27] embeds the GBM into the decoder to extract endmembers and estimate second-order scattering interactions, treating interaction coefficients as trainable network parameters. Based on a post-nonlinear form, the work in [42] approximates the nonlinear term implicitly via FC layers. In contrast, Shahid and Schizas [43] introduce PPNMM-AE, which explicitly models the nonlinear coefficients of PPNMM as learnable parameters. Similarly, Jin and Yang [28] integrate a PPNMM-based

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decoder with superpixel-based graph convolutional networks to capture global spatial information, with nonlinear coefficients also learned as network parameters.

Despite these advancements, existing nonlinear unmixing networks face significant limitations, particularly concerning BMMs. Implicit approaches, especially those relying on FC layers for nonlinear mapping (e.g., [26], [42]), deviate, to some extent, from original mixing model formulations. Consequently, they lack direct interpretability and provide no explicit parameter quantifying the degree of nonlinearity at the pixel level. Furthermore, while some methods explicitly model nonlinear coefficients, they predominantly treat them as network parameters (e.g., [27], [28], [43]), rather than learning them as dynamic features derived from each input pixel (akin to abundance). This fundamentally restricts their adaptability and inference capability: these trained networks cannot dynamically infer corresponding pixel-wise nonlinear coefficients (which are fixed network parameters), preventing them from directly performing unmixing for new pixels, even from the same image. Such inherent rigidity severely limits their flexibility for diverse datasets, as it often necessitates significant redesign and reconfiguration of the network to adapt to varying input data characteristics.

To address these limitations, we propose the dilated transformer-based unmixing network for nonlinear HU (DTU-Net), a novel unsupervised nonlinear unmixing framework. It integrates a dual-branch dilated-transformer encoder for effective multilevel spatial and spectral feature extraction, with a PPNMM-based decoder for accurate linear/nonlinear unmixing. The main contributions are summarized as follows.

1) We propose a spatial branch employing a two-level strategy to capture multiscale and long-range dependencies: integrating multiscale dilated attention (MSDA) for fine-grained features and cascaded downsampling with MHSA for coarse-grained global context.
2) We design a concurrent spectral branch using 3-D-CNNs with channel attention. By capturing subtle spectral discrepancies, this branch complements spatial modeling and addresses the deficiency of fixed spectral grouping used in transformer-based unmixing networks.
3) We develop a PPNMM-based decoder that explicitly models nonlinearity by learning mixing coefficients as pixel-wise features. This distinguishes DTU-Net from existing BMM-based unmixing networks, offering direct pixel-level interpretability and enhancing model adaptability across scenes.

The rest of the article is organized as follows. Section II reviews the relevant unmixing model and attention mechanisms. Section III presents the multiscale dilated transformer network for nonlinear unmixing. Experimental results are presented in Section IV, and Section V concludes the article.

II. Related Works

We begin by briefly reviewing the mixing model used in this article, followed by an introduction to the MSDA mechanism.

A. Polynomial Post-Nonlinear Mixing Model

The PPNMM is a widely used model for nonlinear unmixing, capturing up to second-order interactions between endmembers [16]. Let $\mathbf{y} \in \mathbb{R}^{L\times 1}$ represent the observed spectrum of a pixel across L spectral bands. The PPNMM is expressed as a linear mixing term augmented by a polynomial nonlinear term, which is given by

$$\mathbf{y} = \mathbf{M}\mathbf{a} + b\,(\mathbf{M}\mathbf{a}) \odot (\mathbf{M}\mathbf{a}) + \mathbf{n} \tag{1}$$

where $\odot$ is the Hadamard product, $\mathbf{M} = [\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_R] \in \mathbb{R}^{L\times R}$ is the endmember matrix composed of R endmembers, $\mathbf{a} = [a_1, a_2, \ldots, a_R]^\top \in \mathbb{R}^{R\times 1}$ is the abundance vector, b is a coefficient that balances linear and bilinear effects, and $\mathbf{n} \in \mathbb{R}^{L\times 1}$ represents the additive Gaussian noise. The original PPNMM [16] is a supervised unmixing method, where the endmember matrix $\mathbf{M}$ is known a priori. Due to physical interpretation, both the abundance non-negative constraint (ANC) and sum-to-one constraint (ASC) are imposed, with $a_r \geq 0$ and $\sum_{r=1}^{R} a_r = 1$.

According to (1), the pixel-wise nonlinear effects are characterized by a single scalar parameter b, resulting in a simpler model compared to other second-order nonlinear unmixing models [14], [15]. Notably, when $b = 0$, PPNMM reduces to LMM, which is given by

$$\mathbf{y} = \mathbf{M}\mathbf{a} + \mathbf{n}. \tag{2}$$

As a result, the PPNMM is a flexible model to model both linear and nonlinear mixing scenarios.

B. Multiscale Dilated Attention

The MSDA, proposed in DilateFormer [44], is designed to efficiently capture multiscale semantic information while reducing redundancy in the attention mechanism. Unlike traditional self-attention mechanisms, which focus on locality in shallow layers, MSDA incorporates both locality and sparsity to aggregate information from different scales without introducing significant computation overhead.

1) Sliding Window Dilated Attention: Analogous to the self-attention in transformers [30], the core of MSDA is based on the sliding window dilated attention (SWDA), where keys and values are sparsely selected in a sliding window surrounding the query, followed by self-attention performed on these selected key/value pairs. Let $\mathbf{Q}, \mathbf{K}$, and $\mathbf{V}$ represent the query, key, and value matrices, respectively, where each row is a feature vector of dimension d. The output $\mathbf{X}$ from SWDA is given by

$$\mathbf{X} = \mathrm{SWDA}\,(\mathbf{Q}, \mathbf{K}, \mathbf{V}, r) \tag{3}$$

where r is the dilation rate controlling the sparsity of the attention mechanism. For a given location (i, j) in the input feature map, the corresponding output vector is computed as

$$\mathbf{x}_{(i,j)} = \mathrm{Softmax}\left(\frac{\mathbf{q}_{(i,j)}\mathbf{k}_r^\top}{\sqrt{d}}\right)\mathbf{v}_r \tag{4}$$

where $\mathbf{q}_{(i,j)}$ is the corresponding query vector in $\mathbf{Q}$, and $\mathbf{k}_r$ and $\mathbf{v}_r$ are the key and value vectors sparsely selected from $\mathbf{K}$ and $\mathbf{V}$, respectively, according to the dilation rate r. The selection of $\mathbf{k}_r$ and $\mathbf{v}_r$ is based on the coordinate system

$$\left\{(i', j') \,\middle|\, i' = i + p \times r, j' = j + q \times r\right\}, \; -\frac{w}{2} \leq p, q \leq \frac{w}{2} \tag{5}$$

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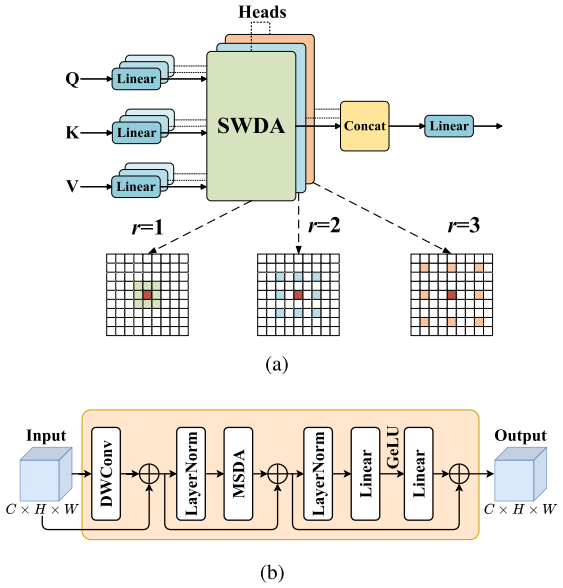

Fig. 1. Illustration of MSDA [44]. (a) MSDA mechanism with three heads. (b) Structure of an MSDA block.

where w is the window size centered at coordinate (i, j), and (p, q) are the relative shifts within the window defined by the dilation rate r. This selective process enables SWDA to capture spatially sparse attention across its receptive field, which facilitates the learning of both local and long-range dependencies and significantly reduces the computational cost compared to dense self-attention.

2) MSDA Block: The core of the MSDA block is the MSDA mechanism, which efficiently aggregates multiscale spatial information by leveraging different dilation rates across various attention heads. Similar to the MHSA mechanism in traditional transformers [30], the feature map is divided into multiple heads along the channel dimension before computing multiscale attention. For each head, SWDA is computed with varying dilation rates. This mechanism is illustrated in Fig. 1(a) and is formulated as

$$\mathbf{h}_i = \text{SWDA}\left(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i, r_i\right), \quad i = 1, \ldots, m \tag{6}$$

$$\mathbf{X} = \text{Linear}\left(\text{Concat}\left[\mathbf{h}_1, \ldots, \mathbf{h}_m\right]\right) \tag{7}$$

where m is the number of heads, $\mathbf{Q}_i$, $\mathbf{K}_i$, and $\mathbf{V}_i$ represent the query, key, and value slices of the feature map for the ith head, respectively, $\mathbf{h}_i$ is the output of ith head, and $\mathbf{X}$ is the output of MSDA that integrates attention information from multiple scales. The strength of MSDA for unmixing lies in its sparse and adaptive attention mechanism. Unlike conventional self-attention (such as MHSA), which suffers from quadratic computational complexity, and fixed-window attention blocks [36], [37], [38], [39], which use a uniform receptive field across all attention heads within an attention block, MSDA's varying dilation rates allow each head to capture features at a different scale. This provides a flexible receptive field that aligns with the diverse spatial structures inherent to HSIs, benefiting accurate abundance estimation.

The structure of an MSDA block is illustrated in Fig. 1(b). The block begins with a 3×3 depth-wise convolution (DWConv) with zero padding, which serves as a conditional position embedding (CPE). Next, the MSDA mechanism captures multiscale spatial dependencies. The output is then processed by a feed-forward network consisting of two linear layers with a Gaussian error linear unit (GeLU) activation in between. Residual connections are applied after each submodule to facilitate information flow, while the feature map dimensions remain unchanged throughout the block.

III. Proposed Method

We propose DTU-Net for nonlinear unmixing, which is primarily based on the PPNMM mechanism described in Section II-A. In the proposed network, the full image serves as the input for unsupervised unmixing, jointly estimating the endmembers, abundances, and nonlinear coefficients for all pixels. As a result, we first extend the pixel-wise PPNMM in (1) to the tensor-based formulation that considers the entire HSI as a whole, preserving its inherent spectral–spatial structure.

Let tensor $\mathcal{Y} \in \mathbb{R}^{n_{\text{row}} \times n_{\text{col}} \times L}$ represent the observed HSI. The tensor-based formulation of PPNMM in (1) is given by

$$\mathcal{Y} = \mathcal{A} \times_3 \mathbf{M} + \mathcal{B} \odot (\mathcal{A} \times_3 \mathbf{M}) \odot (\mathcal{A} \times_3 \mathbf{M}) + \mathcal{N} \tag{8}$$

where $\mathbf{M} \in \mathbb{R}^{L \times R}$ is the endmember matrix, and $\mathcal{A} \in \mathbb{R}^{n_{\text{row}} \times n_{\text{col}} \times R}$, $\mathcal{B} \in \mathbb{R}^{n_{\text{row}} \times n_{\text{col}} \times 1}$, and $\mathcal{N} \in \mathbb{R}^{n_{\text{row}} \times n_{\text{col}} \times L}$ represent the abundance tensor, the nonlinear coefficient tensor, and the additive noise tensor, respectively. The operation $\times_3$ denotes mode-3 multiplication. Broadcasting is used to align the first dimensions of $\mathcal{B}$ and $(\mathcal{A} \times_3 \mathbf{M})$, enabling element-wise multiplication across the spatial dimensions while expanding $\mathcal{B}$ to match the third dimension L.

Primarily, we design the DTU-Net to solve the following optimization problem:

$$\begin{aligned} \underset{\mathbf{M} \geq 0, \mathcal{A} \geq 0, \mathcal{B}}{\arg\min} \; & \|\mathcal{Y} - \mathcal{A} \times_3 \mathbf{M} - \mathcal{B} \odot (\mathcal{A} \times_3 \mathbf{M}) \odot (\mathcal{A} \times_3 \mathbf{M})\|_F^2 \\ & + \mathcal{R}(\mathcal{A}) + \mathcal{R}(\mathcal{B}) \\ \text{s.t.} \; & \sum_{r=1}^{R} \mathcal{A}_{i,j,r} = 1 \quad \forall i \in [1, n_{\text{row}}], \; j \in [1, n_{\text{col}}] \end{aligned} \tag{9}$$

where $\|\cdot\|_F$ denotes the Frobenius norm. Both the abundance ANC and ASC constraints are enforced, and the endmembers are constrained to be nonnegative. The terms $\mathcal{R}(\mathcal{A})$ and $\mathcal{R}(\mathcal{B})$ represent the regularization effects implicitly enforced by the network structures (e.g., convolutional layers and transformers) on the abundance and nonlinear coefficient tensors, respectively. The number of endmembers R is assumed to be estimated a priori, as in [45] and [46].

A. Network Overview

The architecture of the proposed DTU-Net is illustrated in Fig. 2. Building upon the optimization problem in (9), the network consists of two main components: the dilated transformer encoder and the PPNMM-based decoder. The encoder includes two branches: one dedicated to multiscale spatial feature extraction and the other to spectral feature extraction. The features from both branches are fused and

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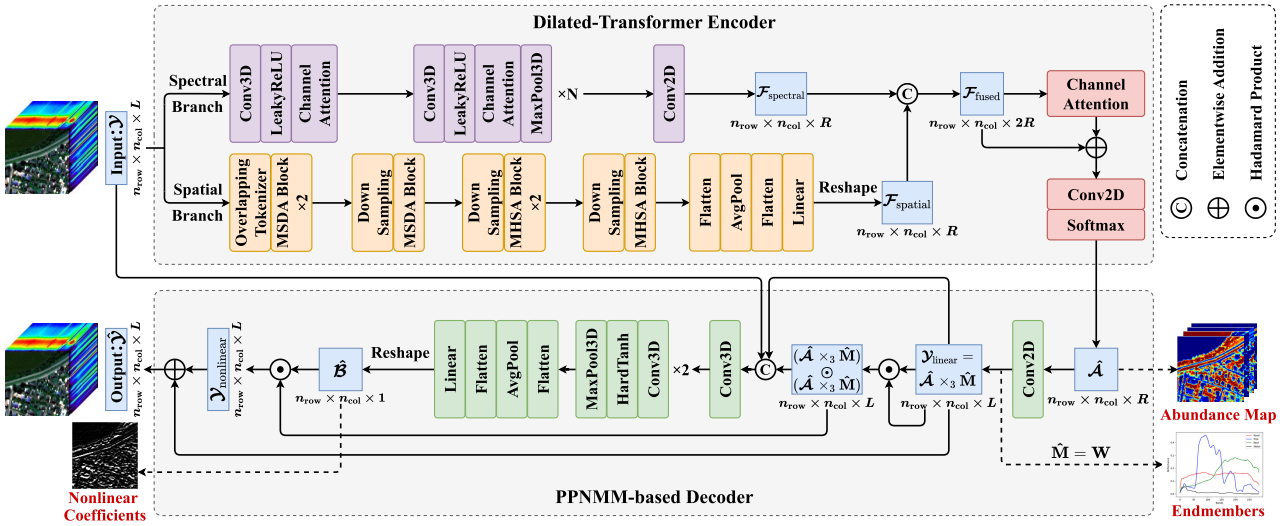

Fig. 2. Overview of the proposed multiscale DTU-Net. DTU-Net consists of two main components: a dilated-transformer encoder and a PPNMM-based decoder. The encoder leverages two distinct branches to extract multiscale spatial and spectral features from the input HSI, which are then fused to generate the abundance tensor. The decoder applies a 2-D convolutional layer to upsample the abundance tensor, whose weight matrix represents the endmember matrix and corresponds to the linear mixing component. The nonlinear coefficient tensor is learned by concatenating and transforming relevant features. Finally, the HSI is reconstructed by combining the linear and nonlinear mixing components using learned features and network parameters, following the PPNMM mechanism. We illustrate the algorithm using the Apex dataset and show examples of estimated endmembers, abundance maps, and nonlinear coefficients.

subjected to dimensionality reduction to generate the estimated abundances. The decoder primarily employs the PPNMM mechanism to reconstruct the HSI. It first applies a convolutional layer to upsample the estimated abundances, with the weights representing the endmember matrix. The nonlinear coefficients are learned as pixel-wise features.

B. Dilated-Transformer Encoder

1) Multiscale Spatial Feature Extraction Branch: The spatial feature extraction branch follows a pyramid structure and consists of five modules. As noted in [44], shallow layers in ViTs exhibit significant sparsity and locality. To address these issues, the first two modules employ MSDA blocks, as illustrated in Fig. 1(b), which are specifically designed to capture multiscale spatial dependencies within a local context. The subsequent two modules use traditional MHSA blocks, which replace the MSDA mechanism with MHSA within the MSDA block, to compute global attention after downsampling. The final module performs feature transformation. This progressive design enables the network to learn from fine-grained local details to broad contextual information. The MSDA blocks used in this branch adhere to the following design: the feature map channels are split into three equal groups, each processed by attention heads with dilation rates $r = 1, 2, 3$. As shown in Fig. 1(a), these correspond to sparsely attended fields of size 3×3, 5×5, and 7×7, respectively. By assigning different dilation rates across channel groups, MSDA enables flexible, multiscale spatial feature fusion within the same block, allowing fine-grained, mid-range, and broader contextual information of feature maps captured concurrently.

In the first module, we use an overlapping tokenizer [47] for patch embedding, which consists of multiple 3×3 convolutional layers with zero-padding. The output, $\mathcal{T}_1$, has a channel dimension C that is data-specific. Multiscale sparse attention is then computed by applying two MSDA blocks, where the subscript k of the operator MSDA_k denotes the number of attention heads. For this module, we use $k = 3$ heads. The complete operations for this module are expressed by

$$\mathcal{T}_1 = \text{OverlappingTokenizer}(\mathcal{Y}) \tag{10}$$

$$\mathcal{T}_2 = \text{MSDA}_3(\mathcal{T}_1), \quad \mathcal{F}_1 = \text{MSDA}_3(\mathcal{T}_2) \tag{11}$$

where $\mathcal{F}_1$ denotes the output of the first module.

In the second module, a convolutional layer with a 3×3 kernel and a 2×2 stride is first used to perform downsampling, integrating features from adjacent pixels. The downsampled feature map is then passed through an MSDA block with six heads (denoted as MSDA_6), where multiscale attention is computed. The operations in this module are expressed as

$$\mathcal{T}_3 = \text{Downsample}(\mathcal{F}_1), \quad \mathcal{F}_2 = \text{MSDA}_6(\mathcal{T}_3) \tag{12}$$

where $\mathcal{F}_2$ is the output of the second module.

In the third and fourth modules, the same downsampling operation was carried out. With the feature map size reduced to 1/16 of its original dimensions, the computational cost of global attention is significantly reduced. This allows us to substitute the MSDA block with an MHSA block, enabling the network to capture global context efficiently. Specifically, the third module employs two MHSA blocks with 12 heads each, while the fourth module uses a single MHSA block with 24 heads. The operations for these two modules are given by

$$\mathcal{T}_4 = \text{Downsample}(\mathcal{F}_2) \tag{13}$$

$$\mathcal{T}_5 = \text{MHSA}_{12}(\mathcal{T}_4), \quad \mathcal{F}_3 = \text{MHSA}_{12}(\mathcal{T}_5) \tag{14}$$

$$\mathcal{T}_6 = \text{Downsample}(\mathcal{F}_3), \quad \mathcal{F}_4 = \text{MHSA}_{24}(\mathcal{T}_6) \tag{15}$$

where $\mathcal{F}_3$ and $\mathcal{F}_4$ denote the output of the third and fourth module, respectively.

Following the computation of multiscale and global attention, the network effectively integrates local and global

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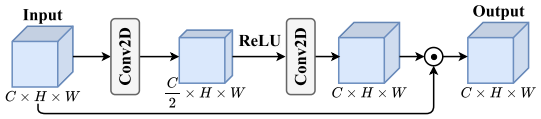

Fig. 3. Architecture of the channel attention module.

features. The output $\mathcal{F}_4$ is processed to generate the spatial feature $\mathcal{F}_{\text{spatial}}$. This process involves a sequence of operations: first, the feature map is flattened along its two spatial dimensions, then an average pooling layer is applied independently to each channel. The resulting channel-wise summary is flattened again into a single vector, passed through a linear layer, and finally reshaped to acquire the spatial feature map $\mathcal{F}_{\text{spatial}}$ with a dimension of $n_{\text{row}} \times n_{\text{col}} \times R$.

2) Spectral Feature Extraction Branch: The spectral feature extraction branch is designed to capture salient spectral features. It begins with an initial 3-D convolutional layer (kernel size $3 \times 1 \times 1$ or $4 \times 1 \times 1$), followed by a LeakyReLU activation and an embedded channel attention mechanism [48]. The 3-D convolution operates along the spectral dimension to extract spectral patterns while preserving spatial locality. This setup performs initial dimensionality reduction on the input image $\mathcal{Y}$. This is followed by a sequence of cascaded blocks. Each block consists of a 3-D convolutional layer (kernel size $3 \times 1 \times 1$ or $4 \times 1 \times 1$) followed by a LeakyReLU activation, a channel attention module, and a 3-D max-pooling layer (window size $2 \times 1 \times 1$), progressively reducing the spectral dimension. Finally, a 2-D convolutional layer with a 3×3 kernel reduces the spectral dimension of the output to the number of endmembers R, resulting in the final spectral feature map $\mathcal{F}_{\text{spectral}}$ which is aligned in size with the abundance tensor.

As shown in Fig. 3, the channel attention module [48] employs a bottleneck structure: a 1×1 convolution reduces channels by half (followed by ReLU), and a subsequent 2-D convolution restores the dimensions. The resulting weights recalibrate the input via element-wise multiplication, effectively highlighting the most discriminative spectral features.

3) Feature Fusion: To comprehensively integrate the spatial and spectral information, $\mathcal{F}_{\text{spatial}}$ and $\mathcal{F}_{\text{spectral}}$ are concatenated along the channel dimension, as follows:

$$\mathcal{F}_{\text{fused}} = \text{Concat}\left[\mathcal{F}_{\text{spatial}}, \mathcal{F}_{\text{spectral}}\right] \tag{16}$$

where $\mathcal{F}_{\text{fused}}$ represents the fused feature map that combines both the multiscale spatial information and the spectral information derived from the original HSI. This fusion ensures that complementary spatial and spectral characteristics are effectively preserved.

To further refine the fused features, we utilize the channel attention module, as shown in Fig. 3, to recalibrate the channel-wise features. The attention-weighted features are then element-wise added to the initial features to enhance spectral representation. Next, a 2-D convolutional layer with a 3×3 kernel and zero-padding is applied to reduce the channel dimensionality, ensuring it matches the number of endmembers. The Softmax function is finally applied to satisfy the abundance ANC and ASC constraints, as given by

$$\hat{\mathcal{A}} = \text{Softmax}\left(\gamma \mathcal{F}\right) \tag{17}$$

where $\mathcal{F}$ denotes the output from the previous processing stage. Here, γ acts as a temperature parameter [49] controlling the sharpness of the Softmax output. Greater γ accentuates dominant components, resulting in an effect analogous to ℓ_1-norm regularization on the abundance maps [37].

C. PPNMM-Based Decoder

The decoder of DTU-Net is designed based on the PPNMM formulation in (8), aiming to reconstruct the HSI while adhering to its physical principles. This subsection details the network design for the linear and second-order nonlinear mixing components.

1) Linear Mixing Component: The decoder receives the estimated abundance tensor $\hat{\mathcal{A}}$ as input. A 1×1 2-D convolutional layer with a stride of 1×1 is used to simulate an FC layer, transforming the abundance tensor from a channel dimension of R to a spectral dimension of L. This operation can be expressed as

$$\mathcal{Y}_{\text{linear}} = \hat{\mathcal{A}} \times_3 \mathbf{W} = \text{Conv2D}\left(\hat{\mathcal{A}}\right) \tag{18}$$

where $\mathbf{W}$ is the weight matrix of the convolution kernel, interpreted as the estimated endmember matrix by $\hat{\mathbf{M}} = \mathbf{W}$. Specifically, the weight matrix $\mathbf{W}$ is enforced to be nonnegative by applying a ReLU activation, such that $\phi(x) = \max(0, x)$ is applied to each element x of $\mathbf{W}$. In this work, $\mathbf{W}$ is initialized using the results from VCA [50].

2) Nonlinear Mixing Component: A major challenge in parameter estimation lies in accurately learning the nonlinear coefficients in model (1). Unlike previous methods [27], [28], [43], which treat pixel-wise nonlinear coefficients as network parameters or use network layers (such as FC layers) to implicitly model nonlinear interactions among endmembers [42], we propose learning these coefficients as features, similar to abundances. This helps to improve network interpretability and stability while adhering to the physical principles of PPNMM.

As described in (9), the estimation of the nonlinear coefficient tensor $\mathcal{B}$ is relevant to three components: the observed HSI $\mathcal{Y}$, the linear mixing part $\mathcal{A} \times_3 \mathbf{M}$, and the second-order interaction term $(\mathcal{A} \times_3 \mathbf{M}) \odot (\mathcal{A} \times_3 \mathbf{M})$. To learn the nonlinear coefficients, these three tensors are concatenated along the channel dimension. This fusion provides a comprehensive representation of the nonlinear interactions by combining raw spectral information, the first-order linear mixing component, and the second-order nonlinear interaction term. The resulting feature map, with an expanded channel size of $3L$, is used as the input for nonlinear coefficient learning.

The feature map is processed through a 3-D convolutional layer (kernel size $5 \times 1 \times 1$) to extract spectral patterns, followed by two cascaded modules. Each module consists of a 3-D convolutional layer of the same kernel size, a HardTanh activation, and a 3-D max-pooling layer (window size $2 \times 1 \times 1$). These modules progressively reduce the

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channel dimension while preserving key features and enabling pixel-wise learning of spectral information pertaining to nonlinear characterization. The resulting feature map is further processed to generate the nonlinear coefficient tensor $\hat{\mathcal{B}}$ by a sequence of operations: first, it is flattened across its two spatial dimensions, then an average pooling layer is applied independently to each channel to summarize its features. The resulting channel-wise summary is flattened again into a single vector, which is then passed through a linear layer. Finally, the output is reshaped to acquire the nonlinear coefficient tensor $\hat{\mathcal{B}}$, with a final dimension of $n_{\text{row}} \times n_{\text{col}} \times 1$.

Using the estimated endmember matrix, abundances, and nonlinear coefficients, the nonlinear mixing component is modeled as

$$\mathcal{Y}_{\text{nonlinear}} = \hat{\mathcal{B}} \odot \left(\hat{\mathcal{A}} \times_3 \hat{\mathbf{M}}\right) \odot \left(\hat{\mathcal{A}} \times_3 \hat{\mathbf{M}}\right). \tag{19}$$

Finally, the reconstructed HSI $\hat{\mathcal{Y}}$ is obtained in accordance with the model in (8), by adding both the linear part (18) and nonlinear mixing part (19).

D. Loss Function

To train the proposed network, two complementary losses are utilized, namely, reconstruction error (RE) loss and spectral angle distance (SAD) loss, which are defined as

$$L_{\text{RE}}\left(\mathcal{Y}, \hat{\mathcal{Y}}\right) = \frac{1}{N}\left\|\hat{\mathcal{Y}} - \mathcal{Y}\right\|_F^2 \tag{20}$$

$$L_{\text{SAD}}\left(\mathcal{Y}, \hat{\mathcal{Y}}\right) = \frac{1}{N}\sum_{i,j} \arccos\left(\frac{\left\langle \mathcal{Y}_{i,j,:}, \hat{\mathcal{Y}}_{i,j,:}\right\rangle}{\left\|\mathcal{Y}_{i,j,:}\right\|_2 \left\|\hat{\mathcal{Y}}_{i,j,:}\right\|_2}\right) \tag{21}$$

where $N = n_{\text{row}} \times n_{\text{col}}$ is the total pixel number of the image.

While the RE loss facilitates accurate pixel-wise reconstruction, it is susceptible to spectral scale variations inherent in measured spectra. To mitigate this, the SAD loss is introduced to measure the angular similarity between the reconstructed and observed spectra. Inherently scale-invariant, it ensures the model learns the correct spectral shape, provided that the estimated endmembers are either free from scaling effects or scaled similarly. By combining these two terms, we leverage L_{RE} for magnitude accuracy and L_{SAD} for spectral shape fidelity, providing a robust training process resilient to spectral scaling effects. The total loss function is a weighted sum of the two losses [32], [37]

$$L = \alpha L_{\text{RE}} + L_{\text{SAD}} \tag{22}$$

where α is a hyperparameter balancing the two losses.

IV. Experiments

In this section, we conduct comprehensive experiments on 18 synthetic datasets, two ray tracing datasets, and three real HSIs to evaluate the unmixing performance of the proposed DTU-Net. For comparison, we selected seven state-of-the-art methods, including the supervised PPNMM [16] and six advanced unmixing networks based on AEs, CNNs, or transformers. Specifically, we considered the CyCU-Net [24], the DeepTrans [32], the Swin transformer unmixing network (Swin-HU) [38], and the UST-Net [37] proposed for linear unmixing, as well as the PPNMM-based AE (PPNMM-AE) [43] and the 3D-CNN-based AE (3D-NAE) for nonlinear mixtures [26]. The proposed DTU-Net is implemented in PyTorch and was trained on a system with an Nvidia RTX 4060 GPU.

To ensure a fair comparison between supervised and unsupervised methods, the results of VCA [50] are used as endmembers for the supervised PPNMM method. For the other unsupervised methods based on deep learning, the endmembers are either initialized using VCA or follow the procedures described in their respective original papers.

The performance of abundance estimation was assessed by the root-mean-square error (RMSE) between the estimated and the true abundance tensor, given by

$$\text{RMSE}_{\text{abun}} = \sqrt{\frac{1}{NR}\left\|\hat{\mathcal{A}} - \mathcal{A}\right\|_F^2}. \tag{23}$$

The endmember estimation was evaluated by the average SAD between the estimated and the true endmember across all endmembers, formulated as

$$\text{SAD}_{\text{end}} = \frac{1}{R}\sum_{r=1}^{R} \arccos\left(\frac{\langle \hat{\mathbf{m}}_r, \mathbf{m}_r\rangle}{\|\hat{\mathbf{m}}_r\|_2 \|\mathbf{m}_r\|_2}\right). \tag{24}$$

When evaluating the nonlinear coefficient estimation for PPNMM-generated datasets, the performance was assessed using the RMSE between the true nonlinear coefficient tensor and the estimated one, defined by

$$\text{RMSE}_{\text{B}} = \sqrt{\frac{1}{N}\left\|\hat{\mathcal{B}} - \mathcal{B}\right\|_F^2}. \tag{25}$$

A. Experiments With Synthetic Datasets

1) Synthetic Datasets Generation: We generated 18 synthetic datasets, each of size 100×100 pixels, by systematically varying four key attributes: number of endmembers, mixing model, noise level, and pixel purity.

We selected $R = 4$ and $R = 8$ spectral signatures from the United States Geological Survey (USGS) digital spectral library, each consisting of $L = 224$ contiguous bands, to serve as endmembers. The corresponding abundance maps were generated using the Hyperspectral Imagery Synthesis tool for MATLAB [51],[1] based on a Gaussian random field to ensure piecewise smoothness and incorporate spatial information. Specifically, to simulate highly mixed cases, we also generated a set of abundance maps for $R = 4$ by enforcing a pixel purity constraint that limited the abundance of any single endmember within a pixel to less than 0.75. This was achieved by proportionally redistributing the excess abundance to other endmembers in the pixel, maintaining the ASC.

Using the reference endmembers and abundance fractions, the datasets were synthesized according to three mixing mechanisms: the LMM in (2), and the bilinear-based GBM [14] and PPNMM [16]. Specifically, the GBM [14] is defined by

$$\mathbf{y} = \sum_{r=1}^{R} a_r \mathbf{m}_r + \sum_{i=1}^{R-1}\sum_{j=i+1}^{R} \beta_{ij} a_i a_j \left(\mathbf{m}_i \odot \mathbf{m}_j\right) + \mathbf{n} \tag{26}$$

[1] http://www.ehu.eus/ccwintco/uploads/f/fb/Synthesis.zip

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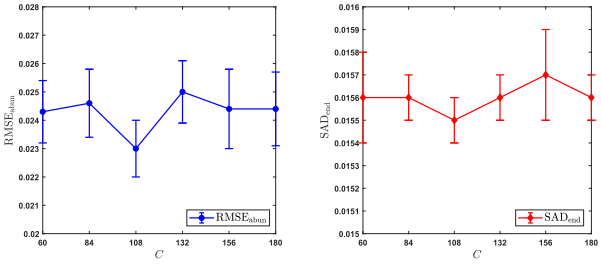
TABLE I
HYPERPARAMETER SETTINGS ON SYNTHETIC DATASETS

	Para. \ Data	SNR = 20 dB			SNR = 30 dB		
		LMM	GBM	PPNMM	LMM	GBM	PPNMM
$R=4$	γ	1	1	1	1	1	1
	α	3	1	1	1	1	5×10^{-1}
	Learning Rate	6×10^{-3} / 10^{-5}	6×10^{-3} / 10^{-4}	1×10^{-2} / 3×10^{-5}	8×10^{-3} / 10^{-5}	1×10^{-2} / 10^{-5}	1×10^{-2} / 10^{-5}
	Weight Decay	10^{-3}	10^{-3}	10^{-3}	10^{-3}	10^{-3}	10^{-3}
$R=4$ <75% purity	γ	8×10^{-1}	8×10^{-1}	1	7×10^{-1}	7×10^{-1}	1
	α	10^{-1}	10^{-1}	2	10^{-1}	10^{-1}	10^{-1}
	Learning Rate	3×10^{-3}/ 2×10^{-2}	3×10^{-3}/ 2×10^{-2}	3×10^{-3} / 3×10^{-2}	3×10^{-3}/ 2×10^{-2}	3×10^{-3}/ 2×10^{-2}	3×10^{-3}/ 2×10^{-2}
	Weight Decay	10^{-3} / 10^{-6}	10^{-3} / 10^{-6}	8×10^{-4} / 10^{-5}	10^{-3} / 10^{-6}	10^{-3} / 10^{-6}	8×10^{-4} / 10^{-5}
$R=8$	γ	1	1	1	1	1	1
	α	3	1	1	3	3	1
	Learning Rate	1×10^{-2} / 10^{-5}	1×10^{-2} / 10^{-4}	1×10^{-2} / 10^{-3}	8×10^{-3} / 10^{-5}	1×10^{-2} / 10^{-5}	1×10^{-2} / 10^{-5}
	Weight Decay	10^{-3}	10^{-3}	10^{-4}	10^{-3}	10^{-3}	10^{-3}



(a) (b)

Fig. 4. Performance metrics on a PPNMM synthetic dataset with SNR = 30 dB as a function of C. The mean of ten runs is plotted for each value, and the error bars represent the standard deviation. (a) $\text{RMSE}_{\text{abun}}$. (b) SAD_{end}.

where $0 \leqslant \beta_{ij} \leqslant 1$ represents the second-order interactions between pairs of distinct endmembers. The values of β_{ij} were sampled from a uniform distribution over the interval $[0, 1]$, following [14]. The PPNMM is defined in (1), where the values of nonlinear coefficient b were uniformly sampled from the interval $[-0.3, 0.3]$, as described in [16].

Finally, Gaussian noise with two signal-to-noise ratios (SNRs), 20 and 30 dB, was added to each dataset to simulate different noise conditions.

2) Hyperparameter Setting: In the experiments on synthetic datasets, the hyperparameters for the proposed DTU-Net were set as follows. Key parameters include the channel dimension C of the embedded feature map in the encoder, the regularization parameter α in the loss function (22), and the scalar γ that controls the sharpness of the Softmax output. In addition, standard neural network hyperparameters such as the number of epochs, learning rate, and weight decay were also considered. The Adam optimizer was used for training, with the number of epochs fixed at 600 in all the experiments. Table I reports the hyperparameters for each dataset, where a slash (/) separates the values applied to the rest of the network and the 2-D convolutional layers, respectively.

The choice of C is crucial as it directly correlates with the number of spectral bands in the HSIs. Setting C too low may result in the loss of essential spectral correlations, while an excessively high value can introduce redundant channel information. We recommend that C not exceed the number of spectral bands, which is 224 in this series of experiments. On the PPNMM-generated dataset with $R = 4$, SNR = 30 dB and unconstrained pixel purity, a candidate set of values [60, 84, 108, 132, 156, 180] was tested. Fig. 4 illustrates the performance metrics, $\text{RMSE}_{\text{abun}}$ and SAD_{end}, as functions of varying C. Based on the results, we set $C = 108$ for all synthetic datasets.

3) Results Analysis: We reported the unmixing results averaged over ten independent runs in Tables II–IV, corresponding to synthetic datasets with $R = 4$ (unconstrained pixel purity), $R = 4$ (highly mixed, <75% pixel purity), and $R = 8$ (unconstrained pixel purity), respectively. Each table contains six datasets, generated by three different mixture models, with two different noise levels. Overall, across all 18 synthetic datasets, the proposed DTU-Net consistently demonstrates superior unmixing performance in both endmember and abundance estimation, ranking first or second among the compared methods.

In terms of endmember extraction, which is evaluated by SAD_{end}, the VCA demonstrates superior performance on synthetic datasets with unconstrained pixel purity and low noise (SNR = 30 dB), particularly when the number of endmembers is small ($R = 4$) (see Table II). This is attributed to its geometric assumption that endmembers correspond to the vertices of a simplex, which holds true when pure pixels are present. However, this advantage significantly diminishes under more challenging scenarios. Under a noise level of SNR = 20 dB, nonlinear unmixing methods, including PPNMM-AE, 3-D-NAE, and the proposed DTU-Net, yield endmember estimations comparable to or slightly better than VCA, especially for GBM and PPNMM datasets. As the number of endmembers increases to $R = 8$, where the unmixing becomes more difficult and the pure pixel assumption less tenable, the proposed DTU-Net consistently outperforms all other methods across the three mixing models and both noise levels, achieving the best endmember estimation as shown in Table IV.

The robustness of the proposed DTU-Net is most pronounced in highly mixed cases (pixel purity <75%). In these scenarios, deep learning-based methods, especially

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TABLE II

UNMIXING RESULTS ON SYNTHETIC DATASETS WITH R = 4, SNR = 20 AND 30 DB, GENERATED USING THE LMM, GBM, AND PPNMM MODELS WITH UNCONSTRAINED PIXEL PURITY, AVERAGED OVER TEN RUNS. BEST AND SECOND-BEST RESULTS ARE MARKED WITH CIRCLED NUMBERS

Unmixing \ Mixing		SNR = 20 dB			SNR = 30 dB		
		LMM	GBM	PPNMM	LMM	GBM	PPNMM
$\mathrm{SAD_{end}}$	VCA	② 0.0065	0.0194	② 0.0202	① 0.0040	① 0.0054	① 0.0193
	CyCU-Net	0.0145	0.0626	0.0845	0.0096	0.0660	0.0631
	DeepTrans	0.0159	0.0334	0.0511	0.0170	0.0351	0.0456
	Swin-HU	0.0105	0.0346	0.0410	0.0118	0.0464	0.0305
	UST-Net	0.0827	0.0822	0.0691	0.0723	0.0701	0.0791
	PPNMM-AE	0.0067	0.0258	② 0.0202	0.0042	② 0.0055	② 0.0202
	3D-NAE	0.0066	② 0.0191	0.0274	0.0131	0.0059	0.0203
	Proposed DTU-Net	① 0.0064	① 0.0183	① 0.0200	② 0.0041	① 0.0054	① 0.0193
$\mathrm{RMSE_{abun}}$	PPNMM	② 0.0171	② 0.0297	② 0.0296	① 0.0066	① 0.0095	② 0.0266
	CyCU-Net	0.1929	0.1737	0.1845	0.1578	0.1649	0.1683
	DeepTrans	0.0239	0.0340	0.0512	0.0191	0.0330	0.0518
	Swin-HU	0.0288	0.0340	0.0428	0.0230	0.0344	0.0421
	UST-Net	0.0782	0.0850	0.0692	0.0866	0.0677	0.0771
	PPNMM-AE	0.0215	0.0325	0.0553	0.0193	0.0224	0.0514
	3D-NAE	0.0649	0.0526	0.0696	0.0410	0.0410	0.0676
	Proposed DTU-Net	① 0.0144	① 0.0258	① 0.0283	② 0.0103	② 0.0146	① 0.0237
$\mathrm{RMSE_B}$	PPNMM			0.1373			0.1713
	PPNMM-AE			② 0.1269			② 0.1652
	Proposed DTU-Net			① 0.1007			① 0.0934

TABLE III

UNMIXING RESULTS ON SYNTHETIC DATASETS WITH R = 4, SNR = 20 AND 30 DB, GENERATED USING THE LMM, GBM, AND PPNMM MODELS WITH <75% PIXEL PURITY, AVERAGED OVER TEN RUNS. BEST AND SECOND-BEST RESULTS ARE MARKED WITH CIRCLED NUMBERS

Unmixing \ Mixing		SNR = 20 dB			SNR = 30 dB		
		LMM	GBM	PPNMM	LMM	GBM	PPNMM
$\mathrm{SAD_{end}}$	VCA	0.1181	0.1228	0.1496	0.1324	0.1229	0.1371
	CyCU-Net	0.0875	0.0981	0.1065	0.0822	0.0840	0.0974
	DeepTrans	① 0.0488	② 0.0560	0.0952	② 0.0583	0.0536	0.0895
	Swin-HU	0.0611	0.0593	0.0893	0.0594	② 0.0528	0.0762
	UST-Net	0.1048	0.1157	0.1258	0.0989	0.1006	0.1036
	PPNMM-AE	0.0730	0.0836	② 0.0778	0.0694	0.0739	② 0.0736
	3D-NAE	0.0973	0.0982	0.1193	0.0918	0.0881	0.1003
	Proposed DTU-Net	② 0.0514	① 0.0512	① 0.0720	① 0.0505	① 0.0501	① 0.0702
$\mathrm{RMSE_{abun}}$	PPNMM	0.0926	0.0940	0.1085	0.1059	0.1034	0.1075
	CyCU-Net	0.2162	0.2812	0.2971	0.1845	0.2159	0.2790
	DeepTrans	0.0465	0.0469	0.0750	0.0383	0.0451	0.0678
	Swin-HU	② 0.0419	② 0.0420	0.0705	② 0.0374	② 0.0403	② 0.0567
	UST-Net	0.0995	0.0927	0.0992	0.0710	0.0767	0.0895
	PPNMM-AE	0.0654	0.0901	② 0.0702	0.0619	0.0838	0.0589
	3D-NAE	0.0771	0.0987	0.0983	0.0705	0.0981	0.0918
	Proposed DTU-Net	① 0.0370	① 0.0377	① 0.0617	① 0.0321	① 0.0363	① 0.0503
$\mathrm{RMSE_B}$	PPNMM			0.1663			0.1052
	PPNMM-AE			② 0.1265			② 0.1022
	Proposed DTU-Net			① 0.1041			① 0.0910

TABLE IV

UNMIXING RESULTS ON SYNTHETIC DATASETS WITH R = 8, SNR = 20 AND 30 DB, GENERATED USING THE LMM, GBM, AND PPNMM MODELS WITH UNCONSTRAINED PIXEL PURITY, AVERAGED OVER TEN RUNS. BEST AND SECOND-BEST RESULTS ARE MARKED WITH CIRCLED NUMBERS

Unmixing \ Mixing		SNR = 20 dB			SNR = 30 dB		
		LMM	GBM	PPNMM	LMM	GBM	PPNMM
$\mathrm{SAD_{end}}$	VCA	② 0.0232	② 0.0255	0.0384	② 0.0077	② 0.0084	② 0.0223
	CyCU-Net	0.0257	0.0729	0.0748	0.0103	0.0506	0.0411
	DeepTrans	0.0376	0.0387	0.0432	0.0109	0.0125	0.0400
	Swin-HU	0.0275	0.0290	0.0364	0.0161	0.0134	0.0357
	UST-Net	0.0920	0.0947	0.0847	0.0908	0.0931	0.0804
	PPNMM-AE	0.0262	0.0289	② 0.0355	0.0092	0.0107	0.0234
	3D-NAE	0.0393	0.0361	0.0595	0.0190	0.0251	0.0467
	Proposed DTU-Net	① 0.0231	① 0.0247	① 0.0298	① 0.0076	① 0.0083	① 0.0221
$\mathrm{RMSE_{abun}}$	PPNMM	0.0349	0.0349	0.0526	② 0.0166	① 0.0173	② 0.0242
	CyCU-Net	0.1915	0.2416	0.2421	0.1275	0.1412	0.1496
	DeepTrans	② 0.0321	0.0342	② 0.0422	0.0275	0.0439	0.0353
	Swin-HU	0.0446	② 0.0328	0.0423	0.0210	0.0235	0.0391
	UST-Net	0.0763	0.0886	0.0751	0.0748	0.0826	0.0711
	PPNMM-AE	0.0391	0.0449	0.0662	0.0355	0.0516	0.0526
	3D-NAE	0.0865	0.0799	0.0918	0.0818	0.0767	0.0862
	Proposed DTU-Net	① 0.0252	① 0.0228	① 0.0328	① 0.0164	② 0.0199	① 0.0229
$\mathrm{RMSE_B}$	PPNMM			② 0.1429			0.1673
	PPNMM-AE			0.1623			② 0.1577
	Proposed DTU-Net			① 0.1042			① 0.1267

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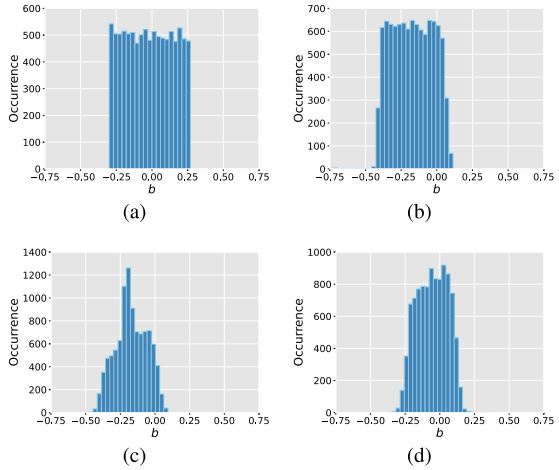

Fig. 5. Histograms of the nonlinear coefficients b obtained by (a) GT, (b) PPNMM, (c) PPNMM-AE, and (d) DTU-Net, on the PPNMM-generated dataset with $R = 4$, SNR = 30 dB, and unconstrained purity.

transformer-based ones, consistently outperform traditional approaches. Notably, DTU-Net achieves the best endmember estimation in 5 out of 6 cases (Table III), validating its superior ability to handle highly mixed scenarios in which the geometric assumptions underlying VCA are violated.

Regarding abundance estimation, evaluated by $\text{RMSE}_{\text{abun}}$, the proposed DTU-Net consistently outperforms other transformer-based and CNN-based methods. This superiority highlights the effectiveness of our encoder in capturing multiscale spatial features and effectively integrating spectral information. Compared to the traditional PPNMM method, DTU-Net provides better abundance estimates in 9 out of 12 cases, particularly when $R = 4$ (pixel purity <75%) and $R = 8$. It is worth noting that supervised PPNMM relies on prior endmember information. Consequently, in scenarios where pure pixels exist, and VCA extracts accurate endmembers (e.g., at SNR = 30 dB or under LMM settings), PPNMM can achieve competitive results. However, as the number of endmembers increases to $R = 8$, DTU-Net, alongside other transformer-based methods like DeepTrans and Swin-HU, demonstrates superior efficacy, especially in noisy cases (SNR = 20 dB). This underlines the robustness of transformer architectures in handling complex datasets. This advantage becomes even more pronounced on highly mixed datasets (<75% pixel purity). In such challenging scenarios, traditional methods exhibit substantial performance degradation, while transformer-based methods, particularly DTU-Net, maintain strong abundance estimation.

On all six datasets generated by the PPNMM model, the nonlinear coefficient estimation of the three PPNMM-derived methods, namely PPNMM, PPNMM-AE, and DTU-Net, is compared using RMSE_B. As observed, DTU-Net, which learns the nonlinear coefficient matrix as a feature map, yields more accurate estimates. Fig. 5 presents the histograms of the nonlinear coefficients estimated by the three PPNMM-based unmixing methods on a PPNMM-generated dataset with $R = 4$, SNR = 30 dB (unconstrained pixel purity). It can be seen that the histogram produced by DTU-Net closely aligns with the ground truth (GT) distribution of b, whereas the other two methods exhibit a bias toward negative values. This further validates the effectiveness of DTU-Net for estimating the nonlinear coefficients.

B. Experiments With Ray Tracing Datasets

1) Data Description: Two datasets of synthetic orchard scenes[2] were generated using a fully calibrated virtual citrus orchard ray tracer [52], [53]. The first consists of citrus trees and a soil background, forming two-endmember mixtures; the second includes additional weed patches, resulting in three-endmember mixtures. These two datasets are referred to as the Orchard dataset with $R = 2$ and $R = 3$, respectively. Each dataset contains 20 × 20 pixels, with the spatial resolution of 2 m. Exact fractional abundance values are available in both datasets. Prior to experiments, 31 water absorption bands (1330–1440 nm and 1770–1950 nm) were removed, leaving $L = 185$ bands for subsequent tasks.

2) Hyperparameter Setting: For the Orchard dataset with $R = 2$, the hyperparameters were set as: $C = 108$, $\gamma = 1$, and $\alpha = 0.1$. The learning rate was 0.01 for the decoder's linear part and 0.02 for the remaining layers, with a weight decay of 0.006 applied to all but the decoder's linear part. For the $R = 3$ dataset, the hyperparameters were $C = 108$, $\gamma = 1$, and $\alpha = 1$, with learning rates of 0.016 and 0.01 for the decoder's linear and remaining layers, respectively. A weight decay of 0.003 was used for all layers except the decoder's linear part. Training was conducted for 600 epochs for both cases.

3) Results Analysis: Quantitative results evaluated by all the compared methods in terms of $\text{RMSE}_{\text{abun}}$ and SAD_{end} on the Orchard datasets with $R = 2$ and $R = 3$ are reported in Table V. These two datasets were generated using a ray-tracing-based simulator, which explicitly models light interactions such as multiple scattering, mutual shading, and structural heterogeneity between tree canopies, soil, and weed components [52]. Given these strong nonlinearities, the proposed DTU-Net demonstrates superior performance in both endmember extraction and abundance estimation on both datasets. Notably, PPNMM-AE, as another deep unmixing method based on PPNMM, also delivers competitive performance on both datasets. In particular, it achieves the second-best abundance estimation result on the more challenging $R = 3$ case, further validating the suitability of PPNMM-based models for handling nonlinear mixing scenarios. Furthermore, transformer-based methods consistently outperform CNN-based and traditional approaches, highlighting their superior ability to capture spatial context in complex and physically realistic scenes. Visual comparisons of the endmember and abundance estimations for the Orchard dataset ($R = 3$) are provided in Figs. 6 and 7. The DTU-Net estimated endmembers closely match the GT spectral signatures, while its abundance maps exhibit a high fidelity to the GT with well-defined boundaries that accurately replicate the spatial distribution of the soil, weed, and tree components. These visual results reinforce the superior quantitative performance reported in Table V.

[2]https://www.biw.kuleuven.be/nonlinear-mixing/scenes_orchard.html

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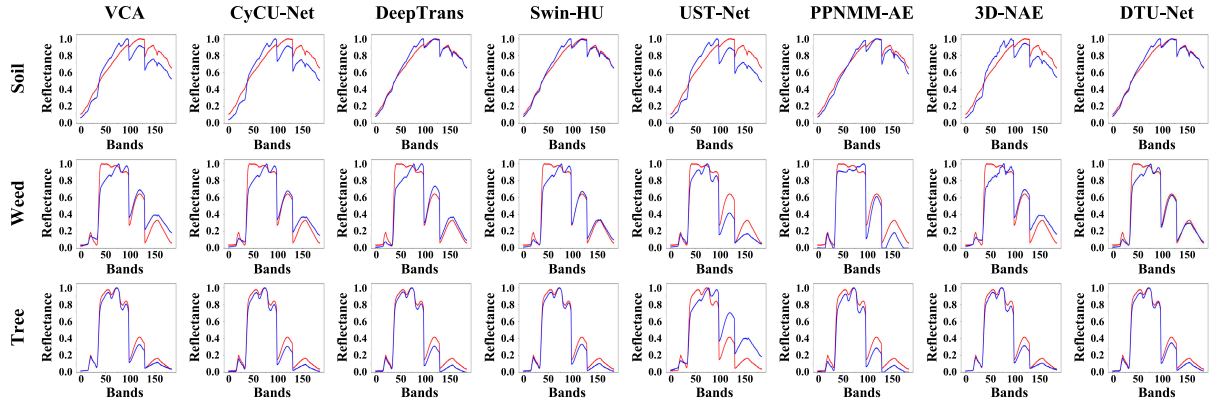

Fig. 6. Endmember estimation for the Orchard dataset with $R = 3$. (Left to right) VCA, CyCU-Net, DeepTrans, Swin-HU, UST-Net, PPNMM-AE, 3-D-NAE, and the proposed DTU-Net. Estimated endmembers are in blue, and GT endmembers are in red.

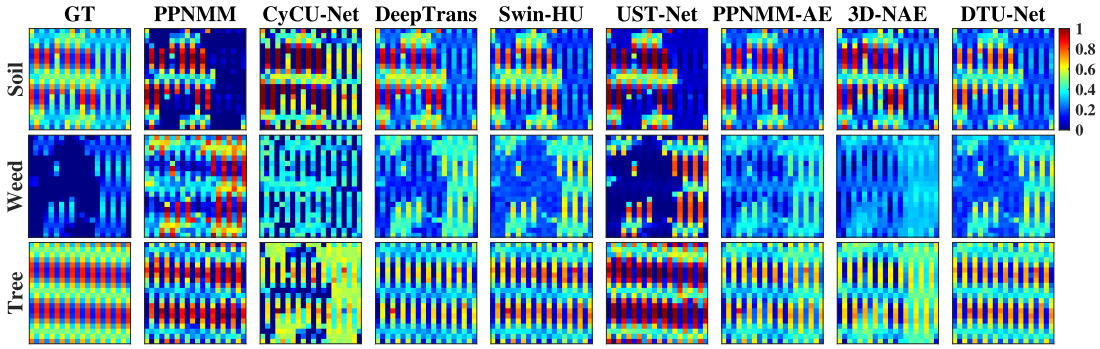

Fig. 7. Abundance map estimation for the Orchard dataset with $R = 3$. (Left to right) GT, PPNMM, CyCU-Net, DeepTrans, Swin-HU, UST-Net, PPNMM-AE, 3-D-NAE, and the proposed DTU-Net.

TABLE V

UNMIXING RESULTS ON RAY TRACING DATASETS AND REAL-WORLD SAMSON, APEX, AND CUPRITE DATASETS. BEST AND SECOND-BEST RESULTS ARE MARKED WITH CIRCLED NUMBERS

	Orchard ($R = 2$)		Orchard ($R = 3$)		Samson		Apex		Cuprite
	SAD_{end}	$\text{RMSE}_{\text{abun}}$	SAD_{end}	$\text{RMSE}_{\text{abun}}$	SAD_{end}	$\text{RMSE}_{\text{abun}}$	SAD_{end}	$\text{RMSE}_{\text{abun}}$	SAD_{end}
VCA-PPNMM	0.0829	0.0816	0.1237	0.2737	0.0666	0.2442	0.3958	0.3179	0.1785
CyCU-Net	0.0965	0.1333	0.1313	0.2392	0.0503	0.1867	0.2768	0.1779	0.1412
DeepTrans	0.0822	② 0.0445	0.1015	0.1586	0.0421	0.1205	② 0.0953	0.1351	0.1258
Swin-HU	② 0.0757	0.0596	② 0.0827	0.1543	0.0374	② 0.0631	0.1156	② 0.1279	0.1326
UST-Net	0.1053	0.0981	0.1314	0.1770	② 0.0358	0.0853	0.0957	0.1529	0.1339
PPNMM-AE	0.0783	0.0682	0.0987	② 0.1410	0.0678	0.2983	0.2890	0.1629	② 0.1233
3D-NAE	0.0880	0.0940	0.1244	0.1578	0.0753	0.2208	0.2958	0.2247	0.1455
Proposed DTU-Net	① 0.0680	① 0.0249	① 0.0706	① 0.1352	① 0.0277	① 0.0540	① 0.0952	① 0.0978	① 0.1121

(a) (b) (c)

Fig. 8. Three HSIs used in the experiments. (a) Samson. (b) Apex. (c) Cuprite.

C. Experiments With Real Datasets

The three real HSIs used in this study are Samson, Apex, and Cuprite, as shown in Fig. 8. It is important to note that the reference GT abundances used to evaluate the unmixing performance of various methods on real datasets are derived under the assumptions of the LMM, with the use of GT endmembers [24], [32]. Despite this, the proposed DTU-Net remains flexible to handle both linear and nonlinear scenarios.

1) Samson: The Samson dataset was acquired by the SAMSON sensor in Monterey Bay in 2006 [54]. For the experiment, a subimage consisting of 95 × 95 pixels was selected, spanning 156 spectral channels. These channels cover the wavelength range of 401–889 nm, with a spectral resolution of 3.13 nm. The Samson imagery comprises three primary endmembers: soil, tree, and water.

The hyperparameters were set as follows: $C = 72$, $\gamma = 1$, and $\alpha = 0.5$. The learning rate was 0.02 for the linear part of the decoder and 0.03 for the remaining layers. A weight

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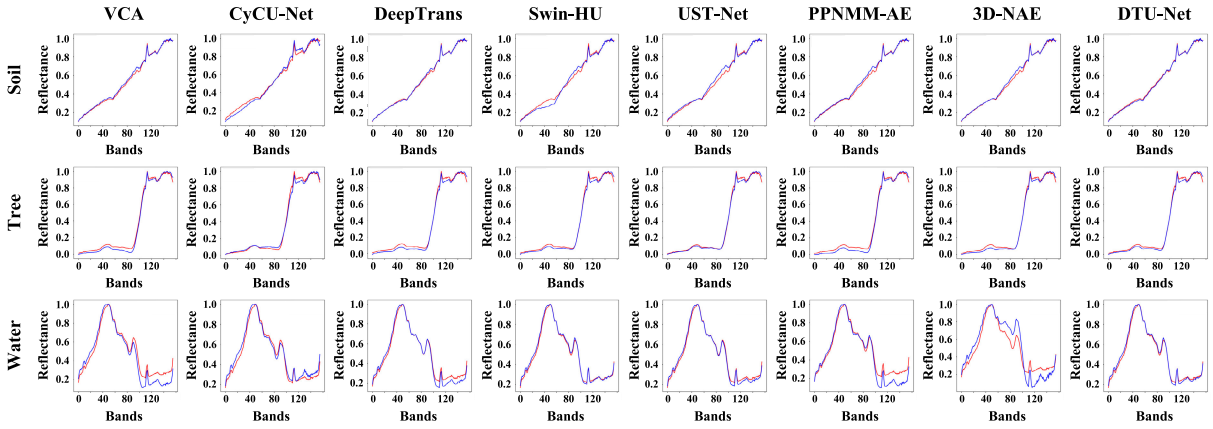

Fig. 9. Endmember estimation for the Samson dataset. (Left to right) VCA, CyCU-Net, DeepTrans, Swin-HU, UST-Net, PPNMM-AE, 3-D-NAE, and the proposed DTU-Net. Estimated endmembers are in blue, and GT endmembers are in red.

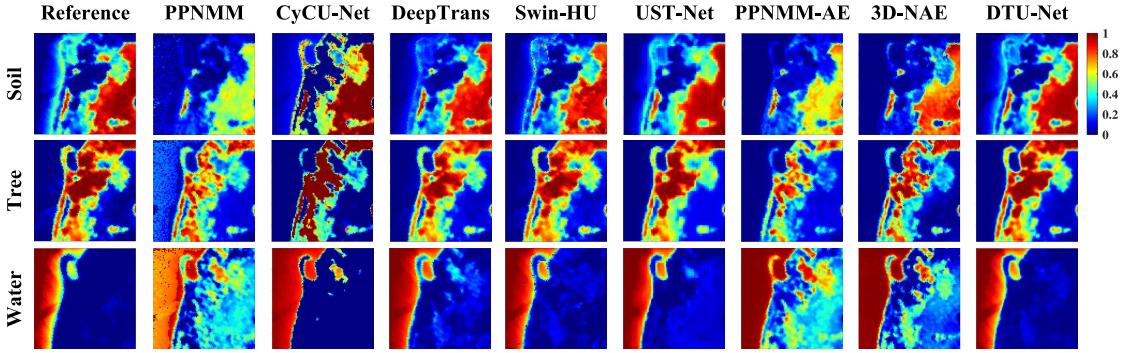

Fig. 10. Abundance map estimation for the Samson dataset. (Left to right) Reference, PPNMM, CyCU-Net, DeepTrans, Swin-HU, UST-Net, PPNMM-AE, 3-D-NAE, and the proposed DTU-Net.

decay of 0.001 was applied to the entire model, and training was conducted over 600 epochs.

The unmixing results for the Samson dataset, evaluated in terms of $\text{RMSE}_{\text{abun}}$ and SAD_{end}, are reported in Table V. The four transformer-based methods, namely, DeepTrans, Swin-HU, UST-Net, and the proposed DTU-Net, generally outperform the other comparative methods. Notably, the proposed DTU-Net demonstrates the best performance for both endmember and abundance estimation. A visual examination of the extracted endmembers and abundance maps in Figs. 9 and 10 reveal that, despite Samson's relatively simple spatial features and few endmembers, DTU-Net provides the most accurate abundance maps compared to the GT. Although UST-Net and Swin-HU achieve good abundance estimation, DTU-Net shows a clear advantage in the abundance map for the "water," providing a clearer delineation of the water's boundary compared to all other methods.

2) Apex: The Apex dataset was acquired by the airborne imaging spectrometer, airborne prism experiment (APEX), during an APEX flight campaign in June 2011 [55]. The image used in the experiment is a cropped image region of 110×110 pixels. It contains $L = 285$ spectral bands ranging from 413 to 2420 nm. This dataset includes four endmembers: water, tree, road, and roof.

The hyperparameters were set as follows: $C = 216$, $\gamma = 1.2$, and $\alpha = 30$. Due to the low accuracy of the initial endmembers extracted by the VCA, a higher learning rate of 0.01 and a lower weight decay of 10^{-8} were applied to the linear part of the decoder. For the remaining layers, the learning rate was set to 0.0016, and the weight decay was set to 3×10^{-4}. Training was conducted over 600 epochs.

Table V presents the unmixing results on the Apex dataset, which poses significant challenges due to its complex spatial–spectral characteristics. The proposed DTU-Net achieves the best performance in both endmember and abundance estimation among all the compared methods. It is followed by DeepTrans in endmember extraction and Swin-HU in abundance estimation, demonstrating the effectiveness of DTU-Net in handling such complex datasets. Visual inspection of Figs. 11 and 12 reveals that estimating the "road" component challenges most methods. While transformer-based approaches generally perform well, DTU-Net yields the most accurate extraction, demonstrating the closest alignment with the GT.

3) Cuprite: The Cuprite dataset was captured by the airborne visible infrared imaging spectrometer (AVIRIS) sensor over the Cuprite mining district in Nevada, USA [23]. It comprises 224 spectral bands ranging from 370 to 2480 nm, with a spatial resolution of approximately 20 m. For analysis, water absorption and low SNR bands are removed, resulting in 187 retained bands. A region of 250×190 pixels is selected from the original image, with a total of $R = 12$ dominant

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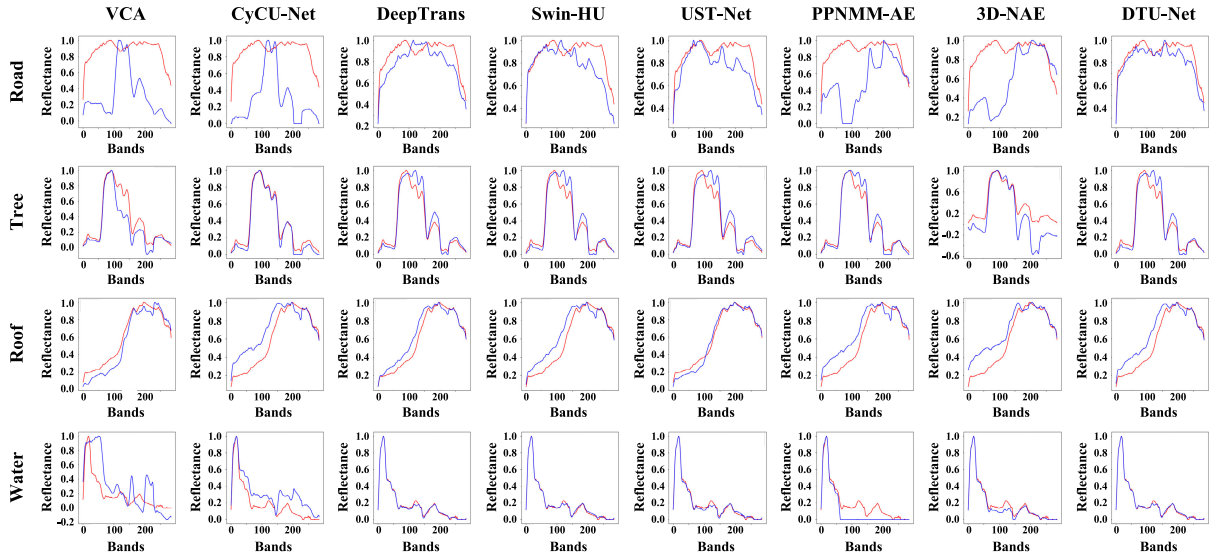

Fig. 11. Endmember estimation for the Apex dataset. (Left to right) VCA, CyCU-Net, DeepTrans, Swin-HU, UST-Net, PPNMM-AE, 3-D-NAE, and the proposed DTU-Net. Estimated endmembers are in blue, and GT endmembers are in red.

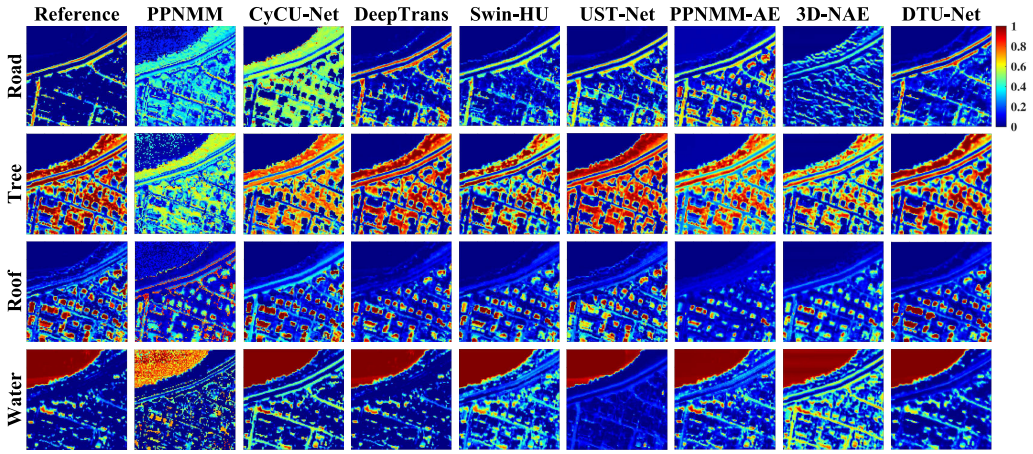

Fig. 12. Abundance map estimation for the Apex dataset. (Left to right) Reference, PPNMM, CyCU-Net, DeepTrans, Swin-HU, UST-Net, PPNMM-AE, 3-D-NAE, and the proposed DTU-Net.

materials considered. While reference GT endmembers were obtained from the USGS Spectral Library, true abundances of the minerals are not available. However, a mineral map generated by the Tricorder software[3] in 1995 serves as a qualitative reference for assessing abundance estimation [56].

In this experiment, we set $C = 120$, $\gamma = 1$, and $\alpha = 5$. The learning rate was set to 0.01 for the entire network. A weight decay of 10^{-4} was applied to all layers of the network except for the decoder's linear part, which was exempted. Training was conducted over 600 epochs.

Table V presents a quantitative comparison of unmixing performance on Cuprite in terms of SAD between the estimated and reference endmembers, while the comparative visualization results for the extracted endmembers and the estimated abundances are shown in Figs. 13 and 14(b), respectively. It can be observed that the endmembers estimated by the proposed DTU-Net most closely match the reference GT, and the estimated abundances are also highly consistent with the reference mineral map.

We visualize the spatial distribution of the estimated nonlinear coefficient tensor $\hat{\mathcal{B}}$ on three real datasets in Fig. 15. The resulting spatial patterns closely align with the texture of the observed images (Fig. 8), indicating that the captured nonlinear distribution is physically meaningful.

D. Ablation Study

1) Ablation on the Dual-Branch Encoder: The encoder of the proposed DTU-Net incorporates a dual-branch structure to extract multiscale spatial and spectral features. To validate the contribution of each branch, we conducted an ablation study on the Samson and Apex datasets, with results presented in Table VI. The results demonstrate that combining both branches yields the best performance, confirming that spatial and spectral features are complementary for accurate unmixing. In addition, when used alone, the spatial branch

[3]http://speclab.cr.usgs.gov/PAPER/tetracorder

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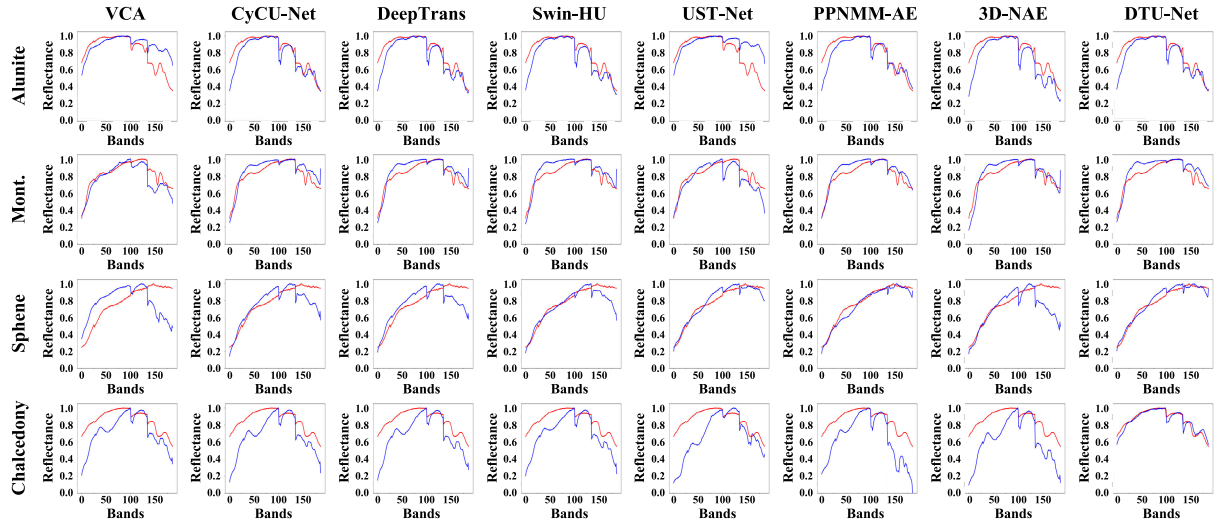

Fig. 13. Endmember estimation for the Cuprite dataset (showing four representative endmembers out of 12). (Left to right) VCA, CyCU-Net, DeepTrans, Swin-HU, UST-Net, PPNMM-AE, 3-D-NAE, and the proposed DTU-Net. Estimated endmembers are in blue, and GT endmembers are in red.

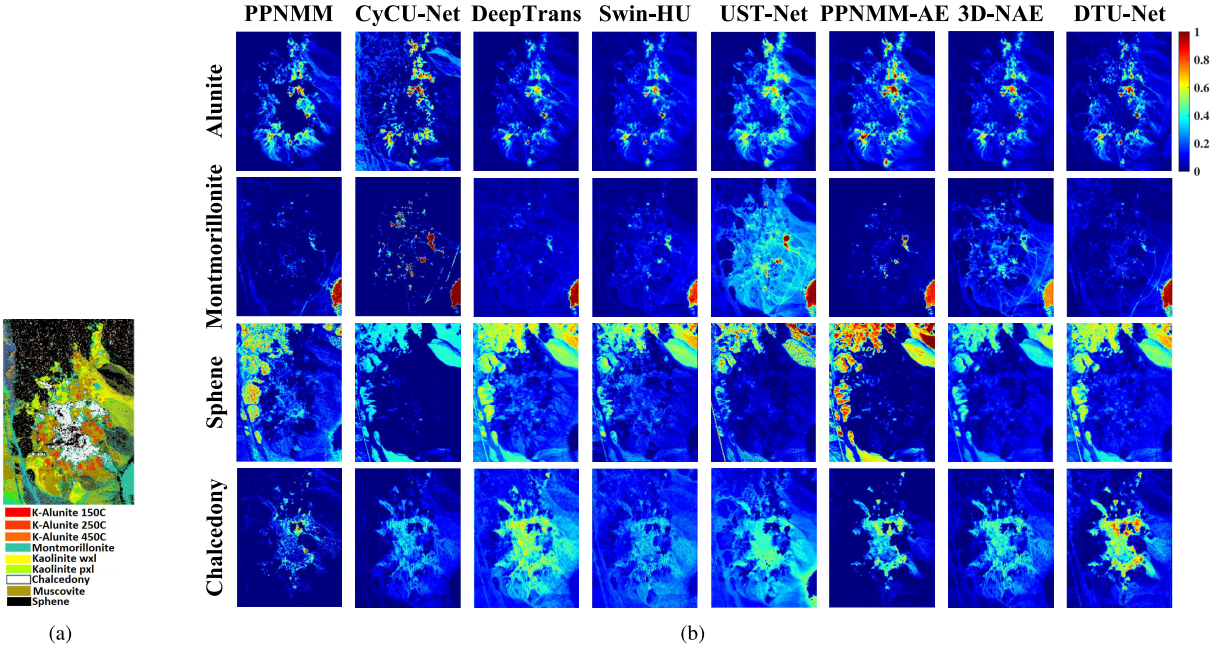

Fig. 14. Reference mineral map and abundance map estimation for the Cuprite dataset. (a) Reference mineral map from Tricorder software. (b) Abundance map estimation corresponding to four representative endmembers (out of 12). (Left to right) PPNMM, CyCU-Net, DeepTrans, Swin-HU, UST-Net, PPNMM-AE, 3-D-NAE, and the proposed DTU-Net.

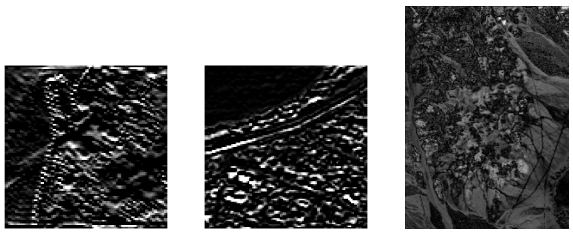

Fig. 15. Spatial distribution of the estimated nonlinear coefficients obtained by DTU-Net on (a) Samson, (b) Apex, and (c) Cuprite.

outperforms the spectral branch, highlighting the effectiveness of our multiscale spatial modeling strategy.

TABLE VI

ABLATION RESULTS FOR DUAL-BRANCH ENCODER. BEST RESULTS IN BOLD

Encoder Branch		Samson		Apex	
$\mathcal{F}_{\text{spatial}}$	$\mathcal{F}_{\text{spectral}}$	SAD_{end}	$\text{RMSE}_{\text{abun}}$	SAD_{end}	$\text{RMSE}_{\text{abun}}$
✓	×	0.0728	0.0956	0.1098	0.1944
×	✓	0.1211	0.3908	0.2727	0.3253
✓	✓	**0.0277**	**0.0540**	**0.0952**	**0.0978**

2) Ablation on the MSDA: To validate the effectiveness of the MSDA in capturing multiscale features, we conducted an ablation study. In DTU-Net, we replaced the MSDA blocks in the multiscale spatial feature extraction branch with standard MHSA blocks. The results on the Samson

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TABLE VII
ABLATION RESULTS FOR MSDA. BEST RESULTS IN BOLD

Attention	Samson		Apex	
	SAD_{end}	$RMSE_{abun}$	SAD_{end}	$RMSE_{abun}$
MHSA	0.0607	0.0863	0.0987	0.1435
MSDA+MHSA	**0.0277**	**0.0540**	**0.0952**	**0.0978**

TABLE VIII
ABLATION RESULTS FOR PPNMM-BASED DECODER. BEST RESULTS IN BOLD

Decoder	Samson		Apex	
	SAD_{end}	$RMSE_{abun}$	SAD_{end}	$RMSE_{abun}$
LMM-based	0.0390	0.0771	0.1026	0.1419
PPNMM-based	**0.0277**	**0.0540**	**0.0952**	**0.0978**

and Apex datasets (Table VII) show that MSDA improves both endmember extraction and abundance estimation. These findings underscore the importance of multiscale modeling: by capturing fine details via dilated attention at multiple scales, DTU-Net outperforms the standard MHSA variant, confirming that multiscale spatial dependencies are essential for accurate unmixing.

3) Ablation on the PPNMM-Based Decoder: To evaluate the contribution of our PPNMM-based decoder, we conducted a comparative ablation study against a variant with a standard LMM-based decoder on the Samson and Apex datasets. As shown in Table VIII, the PPNMM-based decoder consistently outperforms its LMM-based counterpart, particularly in abundance estimation. This superiority highlights the importance of modeling nonlinear effects in complex mixing scenarios and confirms the effectiveness of our physics-based decoding strategy for capturing nonlinear interactions.

4) Ablation on the Input Features for Estimating the Nonlinear Coefficients: In DTU-Net, the nonlinear coefficient tensor $\mathcal{B}$ is estimated by leveraging three concatenated input features: the observed image, the linear mixing term, and the second-order interaction term. To evaluate the necessity and contribution of each component, we conducted an ablation study on the Samson and Apex datasets, as well as a PPNMM-generated synthetic dataset (R = 4, <75% pixel purity, SNR = 30 dB), hereafter referred to as "Syn.30 dB." The following four input configurations were evaluated.

1) *Input 1:* The observed image, $\mathcal{Y}$.
2) *Input 2:* The concatenation of the observed image and the linear mixing term, Concat[$\mathcal{Y}, \hat{\mathcal{A}} \times_3 \hat{\mathbf{M}}$].
3) *Input 3:* The concatenation of the observed image and the second-order interaction term, Concat[$\mathcal{Y}, (\hat{\mathcal{A}} \times_3 \hat{\mathbf{M}}) \odot (\hat{\mathcal{A}} \times_3 \hat{\mathbf{M}})$].
4) *Input 4 (Proposed):* The concatenation of all three feature tensors, Concat[$\mathcal{Y}, \hat{\mathcal{A}} \times_3 \hat{\mathbf{M}}, (\hat{\mathcal{A}} \times_3 \hat{\mathbf{M}}) \odot (\hat{\mathcal{A}} \times_3 \hat{\mathbf{M}})$].

As shown in Table IX, the proposed configuration (Input 4) consistently achieves the best performance. A notable degradation occurs when relying solely on the observed image. Furthermore, Input 3 outperforms Input 2, underscoring the importance of modeling second-order nonlinearity for estimating $\mathcal{B}$. These findings confirm that leveraging all three features is essential for accurate unmixing.

TABLE IX
ABLATION RESULTS FOR INPUT FEATURES FOR ESTIMATING NONLINEAR COEFFICIENTS. BEST RESULTS IN BOLD

Input Feature		Input 1	Input 2	Input 3	Input 4
Samson	SAD_{end}	0.0458	0.0357	0.0321	**0.0277**
	$RMSE_{abun}$	0.0925	0.0835	0.0762	**0.0540**
Apex	SAD_{end}	0.1022	0.1009	0.0983	**0.0952**
	$RMSE_{abun}$	0.1843	0.1136	0.1072	**0.0978**
Syn.30 dB	SAD_{end}	0.0894	0.0828	0.0782	**0.0702**
	$RMSE_{abun}$	0.0685	0.0652	0.0561	**0.0503**
	$RMSE_{B}$	0.1663	0.1118	0.1040	**0.0910**

TABLE X
PARAMETER SCALE (MBS) AND RUNNING TIME PER EPOCH (S) OF COMPARING UNMIXING NETWORKS

	Parameter Scale			Running Time		
	Samson	Apex	Cuprite	Samson	Apex	Cuprite
CyCU-Net	0.12	0.18	0.15	0.11	0.13	0.43
DeepTrans	19.01	30.68	230.38	0.02	0.09	0.08
Swin-HU	387.75	685.10	1526.36	0.07	0.11	0.35
UST-Net	0.06	0.34	0.35	0.61	1.17	2.36
PPNMM-AE	0.01	0.01	0.20	0.16	0.19	0.26
3D-NAE	1.98	4.24	7.84	1.36	2.34	8.57
DTU-Net	42.87	316.68	128.45	0.06	0.16	0.24

E. Computational Complexity

Table X compares the parameter size (MB) and average running time per epoch (s) of DTU-Net against six competing networks (CyCU-Net, DeepTrans, Swin-HU, UST-Net, PPNMM-AE, and 3-D-NAE) on the Samson, Apex, and Cuprite datasets. Note that running times are subject to data dimensions and hyperparameter settings.

V. CONCLUSION

This article presents DTU-Net, an end-to-end multiscale dilated transformer-based nonlinear unmixing network, built upon the AE structure. The encoder features a dual-branch architecture: the first is a dilated transformer branch that captures multiscale spatial dependencies using SWDA and MSDA to represent long-range and multiscale spatial correlations. The second branch incorporates 3-D-CNNs and channel attention mechanisms to model spectral dependencies. The fused features from the two branches derive the abundance maps. The decoder, designed based on the PPNMM, first mimics the linear mixture part before extending to the full PPNMM mechanism. By explicitly modeling the relationships between endmembers, abundances, and nonlinear coefficients, the decoder captures both linear and nonlinear mixing characteristics. By learning the nonlinear coefficients as pixel-wise features, the decoder ensures both flexibility and interpretability, enabling accurate and robust unmixing across diverse datasets. Extensive validation on synthetic, ray tracing, and real datasets confirms DTU-Net's superior robustness compared to existing methods. While advanced deep learning techniques, such as transformers, have significantly enhanced abundance estimation, current unmixing networks for endmember estimation primarily rely on the weight matrix of FC layers. Future work will develop more adaptive techniques for endmember estimation.

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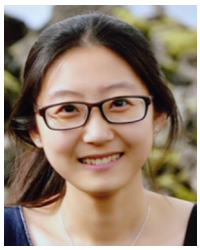

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